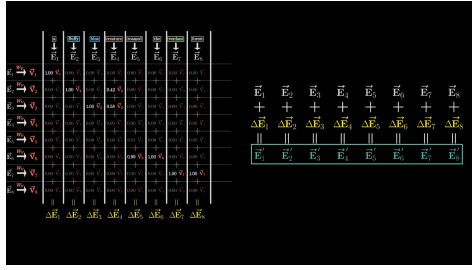
The Transformer Block

Many heads, position, and the residual stream

Where we got to

Last lecture built **one head** of self-attention, end to end: every token emits a query, a key and a value; scores are $QK^\top/\sqrt{d_k}$; softmax turns them into weights; the output is a weighted sum of values.



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That head can already move a word's meaning toward its context. It is also, as we are about to see, **completely blind to word order**.

Three things stand between that and a Transformer

1. **One head is not enough.** A single pattern of who-looks-at-whom cannot capture syntax and meaning and position at once.
2. **Attention cannot see order.** We will prove this, not assert it.
3. **One layer is not a network.** Something has to hold the stack together and let gradients through.

Each of the three has a fix, and the three fixes bolted onto one head of attention *are* the Transformer block. That is the whole lecture.

Outline

Many heads

One pattern of attention is never enough.

Run it h times, in parallel

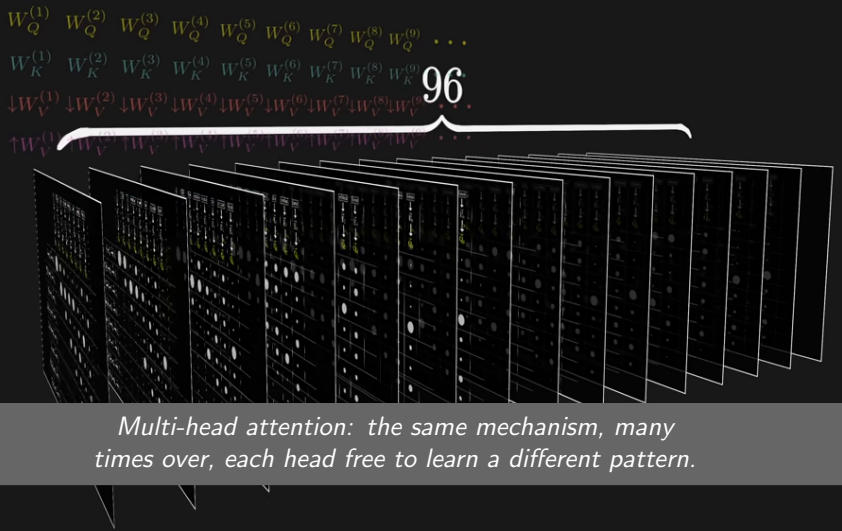
Nothing about one head changes. You simply learn h independent sets of W^Q, W^K, W^V, run them at the same time, concatenate the outputs, and mix them with one more matrix W^O:

$$\begin{aligned} \text{MultiHead}(X) &= \text{Concat}(\text{head}_1, \ldots, \text{head}_h)\, W^O, \\ \text{head}_i &= \text{Attention}(XW_i^Q, XW_i^K, XW_i^V) \end{aligned}$$

The parameter count does not grow. Each head works in a smaller space: $d_k = d_{\text{model}}/h$. With $d_{\text{model}} = 512$ and $h = 8$, every head is 64-dimensional, so each W_i^Q is 512×64. Eight of them is $8 \times 512 \times 64 = \mathbf{262{,}144}$ – exactly one 512×512 matrix. You are not adding capacity, you are **splitting** it into independent views.

And because the heads do not talk to each other until the concat, they are one more big batched matmul – the parallelism argument from last lecture survives intact.

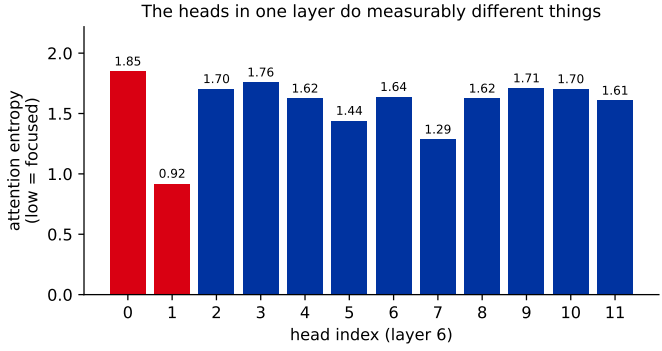
Multi-headed attention



Multi-head attention: the same mechanism, many times over, each head free to learn a different pattern.

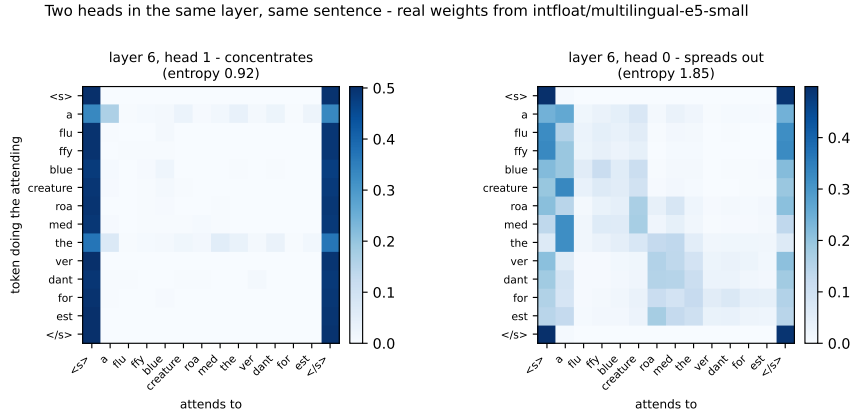
Do heads actually specialise, or is that a story?

Easy to claim, so let us measure it. Take a real pretrained encoder, feed it one sentence, and pull out the actual attention weights from every head in one layer.



Attention entropy $= -\sum_j a_{ij} \log a_{ij}$, averaged over rows: low means a head concentrates on a few tokens, high means it hedges across many.

Two heads, same layer, same sentence



Head 1 concentrates (entropy **0.92**); head 0 spreads (entropy **1.85**) – in the *same* layer, on the *same* sentence, with the same input. Different views is not a metaphor; it is what the weights do.

What the specialisations look like

Across models and layers, heads reliably fall into recognisable jobs:

Kind of head	**What it attends to**
positional	the previous token, or the next one
syntactic	the verb belonging to this subject; the noun this adjective modifies
coreference	the name a pronoun refers to
lexical	the other half of a multi-token name (*Harry* $\leftrightarrow$ *Potter*)

Read these labels with some suspicion. Nobody assigned these jobs; they are patterns *we* recognise afterwards, and plenty of heads do nothing so tidy. The measured claim is the one on the last slide: the heads differ. The names are our interpretation.

Position

Attention has no idea which word came first.

Predict first

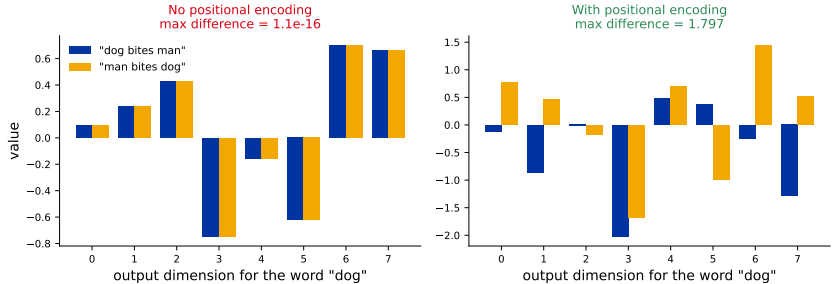
Take the same three word vectors in two orders:

“dog bites man” vs **“man bites dog”**

Does one head of self-attention give the word “dog” a different output vector?

Commit to an answer.

No. Not slightly different – identical.



Left: the largest difference across the whole output vector is $\mathbf{1.1 \times 10^{-16}}$ – floating-point zero. Self-attention is **permutation-equivariant**: permute the inputs and the outputs come back in the same permuted order, unchanged. (Not *invariant* – the outputs do move; each one just stays attached to its own token.) A bag of words and a sentence are the same object to it.

Why that had to be true

Look back at the mechanism and there is nowhere for order to enter:

- $\mathbf{q}_i \cdot \mathbf{k}_j$ depends on the two *vectors*, not on i or j.
- The softmax is over a set of scores.
- The output is a *sum*, and addition does not care about order.

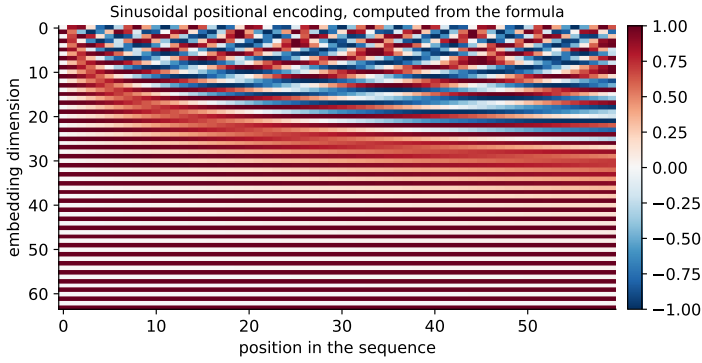
Concretely: in *both* sentences, "dog" attends to the same three vectors {dog, bites, man} and gets the same three weights. Only the order in which we *listed* them changed, and the sum does not notice.

An RNN got order for free – it processed tokens one at a time, so position was implicit in the computation. Attention threw the recurrence away to gain parallelism (L24), and **order was what it paid**. So order has to be put back by hand.

The fix is almost crude: add a position-dependent vector to each embedding before the first block, so that "dog in slot 1" and "dog in slot 3" are different inputs.

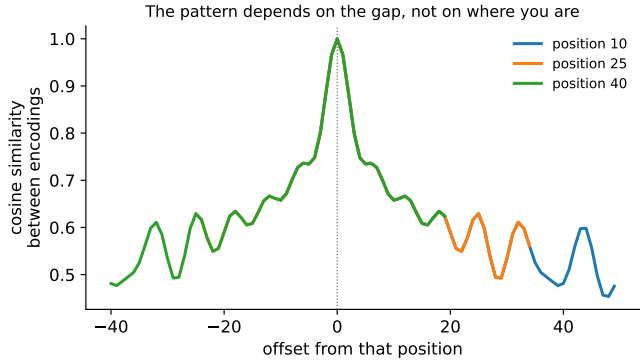
Sinusoidal positional encoding

$$PE_{(pos,\,2i)} = \sin\left(\frac{pos}{10000^{2i/d_{\text{model}}}}\right) \qquad PE_{(pos,\,2i+1)} = \cos\left(\frac{pos}{10000^{2i/d_{\text{model}}}}\right)$$



Each dimension is a sine wave; dimensions further down the embedding oscillate more slowly. Every position gets a distinct fingerprint, and no learned parameters are involved.

Why sinusoids rather than just the number *pos*



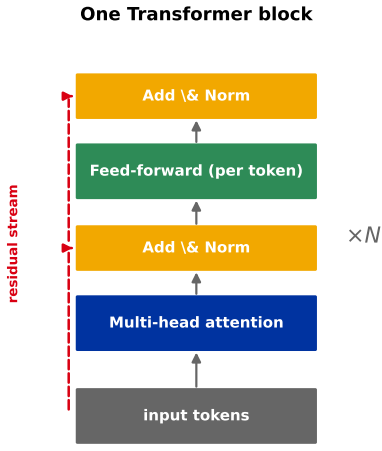
The similarity between two encodings depends on **how far apart** the positions are, not on where they sit. Measured here: the curves for positions 10, 25 and 40 agree to within **0.0000**. So a head can learn "attend three tokens back" once and reuse it everywhere – including at positions longer than anything it was trained on.

Modern models mostly use **RoPE** instead, which rotates **q** and **k** by an angle proportional to position. Same goal, better extrapolation.

The block

What you actually stack.

The Transformer block



schematic

Multi-head attention – tokens exchange information. The *only* place in the whole block where tokens see each other.

Feed-forward network (FFN) – an ordinary MLP (**L14**) applied to each token separately, usually widening 4× and back. Most of the parameters live here.

Add & Norm – the residual connection (**ResNet, L17**) plus LayerNorm: BatchNorm's idea from **L15**, but normalising across each token's *own features* instead of across the batch.

Three familiar pieces from three earlier chapters. The only genuinely new component in a Transformer is the attention – everything else you already met.

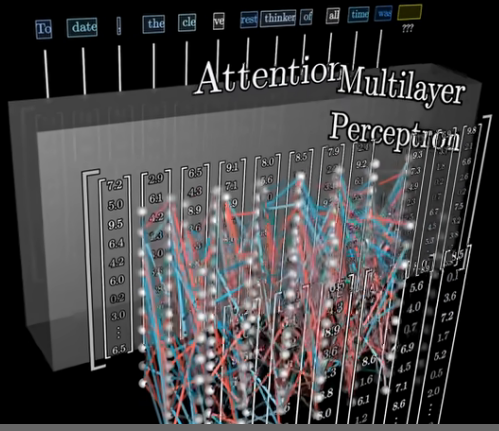
The residual stream is the real object

It is tempting to read the block as "attention, then an MLP". A more useful reading: there is a **stream** of token vectors running straight through the network, and each sublayer *writes an update into it*.

$$\mathbf{x} \leftarrow \mathrm{Norm}\big(\mathbf{x} + \mathrm{Attention}(\mathbf{x})\big) \qquad \text{then} \qquad \mathbf{x} \leftarrow \mathrm{Norm}\big(\mathbf{x} + \mathrm{FFN}(\mathbf{x})\big)$$

Two consequences. **Gradients** reach the early layers unimpeded, which is why you can stack 96 of these – exactly the ResNet argument from L17. And every sublayer's output is a **correction** to a running representation, not a fresh one, which is why the same token's vector can accumulate meaning gradually with depth.

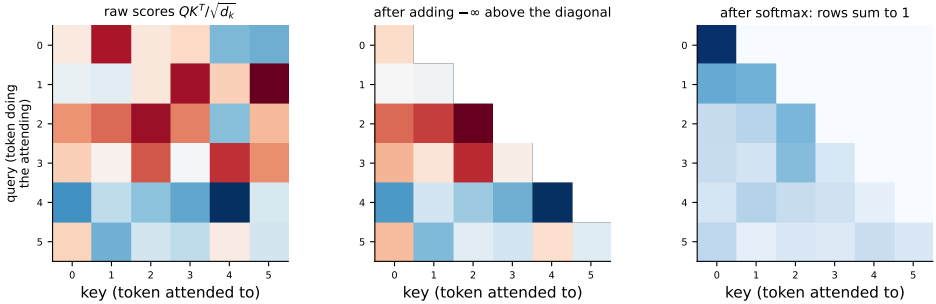
Read the order carefully: the norm comes *after* the addition, matching the diagram and the 2017 paper. Modern models move it **inside**, and we come back to why in three slides.



The flow through a block: attention lets tokens talk, then the MLP processes each one alone. Repeat.

Masking: how a decoder avoids reading the future

A model trained to predict the next token must not be allowed to look at it. The fix is one line: add $-\infty$ above the diagonal *before* the softmax.



$e^{-\infty} = 0$, so those weights become **exactly** zero – measured here as 0.000 attention paid to any future token – and each row still sums to one.

Why that trick matters more than it looks

Without masking, you would have to train on one prediction at a time: feed the first t tokens, predict token $t+1$, repeat.

With masking, a single forward pass over a sequence of length n produces n **training signals at once** – every prefix is being predicted simultaneously, and none of them can cheat.

This is a large part of why pretraining on internet-scale text is affordable at all. The causal mask is not a safety measure; it is a **throughput** decision that happens to enforce correctness.

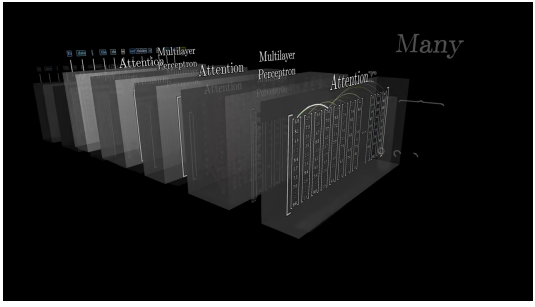
Modern block recipe

The 2017 block has been tuned. Three changes you will meet in every current model:

2017	Now	Why
$\mathrm{Norm}(\mathbf{x} + f(\mathbf{x}))$	$\mathbf{x} + f(\mathrm{Norm}(\mathbf{x}))$	**Pre-LN**: trains stably, no warmup
LayerNorm	**RMSNorm**	cheaper, no mean subtraction
ReLU in the FFN	**SwiGLU**	gated, better quality per parameter

None of these changes the story. If you understand attention, residuals and a per-token MLP, you understand a 2026 model – the rest is engineering that moved a few percent and then stayed.

Now stack it



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Blocks are identical in shape and stacked N deep: 12 for the original BERT, 96 for GPT-3.

Each pass alternates **tokens talking** (attention) with **each token thinking alone** (the MLP).

A token's vector starts as "the word *king*" and, layer by layer, accumulates everything the context implies – who this king is, what happened to him, what the sentence expects next.

Same block, many times. That is the entire architecture.

Recap

- **Multi-head** = the same attention run h times on *splits* of the embedding, then concatenated. No extra parameters, several independent views. Measured: 12 heads in one layer, attention entropy from 0.92 to 1.85.
- **Attention cannot see order** – proved, not asserted: reversing a sentence changed the output by 1.1×10^{-16}. Position must be added by hand.
- **Sinusoidal encodings** give every position a fingerprint whose similarity depends on distance, not absolute location. Modern models use RoPE for the same reason.
- **The block** = multi-head attention, then a per-token MLP, each wrapped in a residual and a norm. Attention is the only new part; residuals are L17 and norms are L15.
- **Causal masking** sets future weights to exactly zero, which turns one forward pass into n training examples.

You can now build one.

What you cannot yet say is why *this* arrangement
beat everything else that was tried -
or what it is still bad at.

Next: the family, how it learns, and the honest limits.